Binomial Expansion

Exam code: 9MA0

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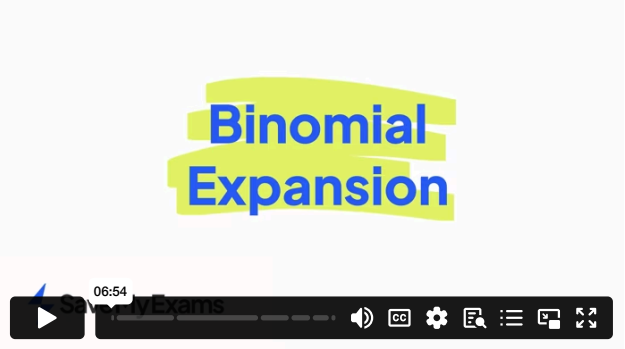

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Binomial expansion

What is the binomial expansion?

$$
\begin{aligned}
(a+b)^n = {} & {}^nC_0 (a)^n (b)^0 \\
& + {}^nC_1 (a)^{n-1} (b)^1 \\
& + {}^nC_2 (a)^{n-2} (b)^2 \\
& + {}^nC_3 (a)^{n-3} (b)^3 \\
& + \dots \\
& + {}^nC_r (a)^{n-r} (b)^r \\
& + \dots \\
& + {}^nC_{n-1} (a)^1 (b)^{n-1} \\
& + {}^nC_n (a)^0 (b)^n
\end{aligned}
$$

$$
{}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} \qquad n \in \mathbb{N}
$$

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- Look at the pattern
  - Start at ${}^nC_0$, then ${}^nC_1$, ${}^nC_2$, etc
  - Powers of **a** start at **n** and decrease by 1
  - Powers of **b** start at 0 and increase by 1
- There are shortcuts but these hide the pattern
  - ${}^nC_0 = {}^nC_n = 1$
  - ${}^nC_1 = {}^nC_{n-1} = n$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- ! means factorial

$$(a+b)^n = a^n + na^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots + b^n$$

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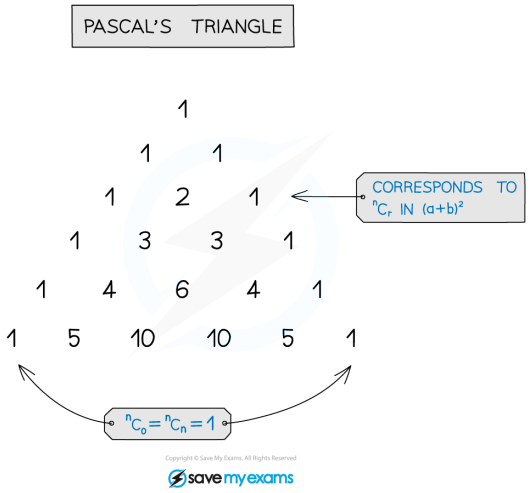
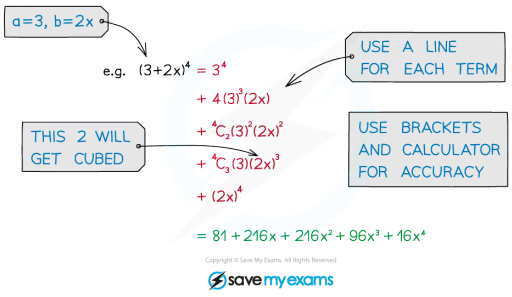
- This is given in the formula booklet

What is Pascal's triangle?



- Pascal's triangle is an alternative to ${}^nC_r$
- It is useful for lower values of **n**
- For larger **n** it is slow and prone to arithmetic errors

How do I expand brackets with the binomial expansion?



- Use a line for each term to make things easier to read and follow
- Use brackets, particularly helpful when negatives involved
- Use a calculator for ${}^nC_r$

Worked Example

Find the first four terms, in ascending powers of x, in the expansion of $(2-3x)^6$

USE SEPARATE LINES AND BRACKETS TO KEEP WORKING CLEAR

$$(2-3x)^6 \approx (2)^6$$
$$+ 6(2)^5(-3x)$$
$$+ {}^6C_2(2)^4(-3x)^2$$
$$+ {}^6C_3(2)^3(-3x)^3$$

TYPE ${}^6C_3 \times (2)^3 \times (-3)^3$ INTO CALCULATOR

$$= 64 - 576x + 2\,160x^2 - 4\,320x^3$$

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Applications of binomial expansion

What are binomial expansions used for?

- Expanding brackets, it is usual to be asked for either ...
  - ... the first few terms only

- or (the coefficient of) a particular term

- **Solving problems** in unknowns

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